

Objective is to find a (prime) factor of  $m$

Define  $d_n = \gcd(2^{n!} - 1, m)$

Observe that  $\underline{d_n \mid d_{n+1}}$   $\left\{ \begin{array}{l} (2^{(n+1)!} - 1) \\ = (2^{n!})^{n+1} - 1 \end{array} \right\}$   
 $\& 2^{n!} - 1$  divides it.

The following facts  
are true.

$$\gcd(a, b) = \gcd(a + bx, b) \quad \text{for } x \in \mathbb{Z}$$

If for some  $n$ ,  $\underline{1 < d_n < m}$ , then we get  
a nontrivial factor of  $m$ .

• We can replace 2 by a  $> 1$ , &  $\underline{n!}$  can also  
 be replaced by a different sequence.

Define  $g_n = \gcd(a_{n-1}, m) \quad \left| \quad g_1 = \gcd(a-1, m) \right.$

where  $a_n = a_{n-1}^{r_{n-1}}$   $\left( \begin{array}{l} \text{take} \\ a_1 = a \end{array} \right)$

where  $r_{n-1}$  a seq<sup>n</sup> of positive numbers.  
 $\left( \begin{array}{l} \text{for } d_n, \quad a_{n-1} = 2^{(n-1)!} \\ \& r_{n-1} = n \end{array} \right)$

Then  $\underline{g_{n+1} = \gcd(\underline{a_1^{r_1} a_2^{r_2} \dots a_n^{r_n}} - 1, m)}$

Observe that  $\underline{g_i \mid g_{i+1} \quad \forall i}$



$$\left[ \text{since } (a^{r_1 r_2 \dots r_{i-1}} - 1) \mid (a^{r_1 r_2 \dots r_{i-1} r_i} - 1) \right]$$

Suppose  $p$  is a prime s.t.  $p \mid m$  &  
 $(p-1) \mid r_1 r_2 \dots r_{n-1}$

Then  $a_n = a^{r_1 r_2 \dots r_{n-1}} = a^{(p-1)t} \equiv 1^t = 1 \pmod{p}$  for some  $n$

So if  $(p-1) \mid r_1 r_2 \dots r_{n-1}$ , then  $p \mid g_n$

Suppose  $\{q_n\}$  is the seq<sup>n</sup> of all prime powers  
 in increasing order so that

$$q_1 = 2, q_2 = 3, q_3 = 4, q_4 = 5, q_5 = 7, q_6 = 8, q_7 = 9, q_8 = 11,$$

$$q_9 = 13, q_{10} = 16, q_{11} = 17, \dots$$

& choose  $r_n$  to be the prime that divides  $q_n$

Then  $r_1 = 2, r_2 = 3, r_3 = 2, r_4 = 5, r_5 = 7, r_6 = 2, r_7 = 3, r_8 = 11$   
 $r_9 = 13, r_{10} = 2, r_{11} = 17, r_{12} = 19$

$$\text{Observe } \text{lcm}(1, 2, \dots, 100) = \text{lcm}(q_1, q_2, \dots, q_{100})$$

$$= \underline{r_1 r_2 \dots r_{100}}$$

More generally



Remark  $\text{lcm}(2, \dots, \underbrace{q_j}_{j}) = \text{lcm}(q_1, q_2, \dots, q_j) = \prod_{i=1}^j r_i$

just st.  $\boxed{q_j \leq l.}$

Since,  ~~$r_1 r_2 \dots r_n \leq (n+1)!$~~

~~$r_i \leq (i+1)$~~

~~$r_i \leq q_i$~~

we shall less calculation with the new choice  
if  $p-1$  has got small prime factors, this method  
works for the new choice.

Algorithm: 1. Choose  $a$  ( $\therefore a=2$ )

2. loop  $j=2, 3, 4, \dots$  up to some bound  $B$

3. set  $a = \underline{a^j \pmod m}$   $a = a^j \pmod m$

4. compute  $d = \underline{\text{gcd}(a-1, m)}$

5. if  $1 < d < m$ , return a proper  
factor  $d$  of  $m$ .

6. otherwise increase  $j$ . (go to 2)

In this algo, we use the ~~addition~~ multiplication.

Structure of  $\underline{\mathbb{F}_p^\times} = \mathbb{F}_p \setminus \{0\}$  (if  $p \nmid m$ )

& one can replace this group by  
an elliptic curve.